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Integrated modeling, surrogate-assisted multi-objective optimization, and decision-making of compact supercritical carbon dioxide Brayton cycle nuclear energy systems for lightweight and high-efficiency applications

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ABSTRACT

Supercritical carbon dioxide Brayton cycle nuclear energy systems are promising candidates for next-generation power generation in mass- and volume-constrained applications, including marine propulsion, aerospace platforms, and advanced small modular reactors. However, comprehensive system-level performance evaluation and optimization studies for direct-cycle supercritical carbon dioxide nuclear energy systems remain limited, and existing multi-objective optimization studies often rely on subjective judgment, lacking an objective and data-driven decision basis. This study proposes an integrated modeling, surrogate-assisted optimization, and data-driven decision-making framework for compact supercritical carbon dioxide Brayton cycle nuclear systems. A validated thermodynamic model and detailed component mass models are developed, with thermal efficiency, specific work, and system mass defined as the key performance indicators. A neural-network-based surrogate model is constructed to accelerate reactor mass estimation under lifetime reactivity constraints, enabling efficient reactor–cycle coupled optimization. A multi-objective optimization is performed using the genetic algorithm, and the objective weight decision method is applied for data-driven objective weighting and ranking of Pareto-optimal solutions. Results reveal that system mass (42.67 %) has the highest influence, followed by thermal efficiency (30.33 %) and specific work (27.00 %) within the range of the set optimization parameters. The optimal configuration achieves a 48.6 % reduction in system mass relative to a conventional efficiency-driven design, while maintaining a thermal efficiency of 42.995 % and specific work of 112.572 kJ/kg. The proposed framework provides a quantitative and transparent decision basis for compact supercritical carbon dioxide nuclear systems, offering practical guidance for other high-power-density energy applications such as mobile and modular reactors.

1. Introduction

As the global energy system landscape shifts towards low-carbon and high-efficiency solutions, Supercritical Carbon Dioxide ($S\text{-}CO_2$) nuclear energy system, which employs $S\text{-}CO_2$ as both the reactor coolant and the power conversion working fluid, has attracted considerable interest due to its superior thermal efficiency, high power density, compact system design, and modularization potential, providing significant advantages over conventional working fluids such as steam and helium gas [1]. These characteristics make $S\text{-}CO_2$ nuclear systems particularly well-suited for a wide range of applications, including marine propulsion [2], space exploration [3], and mobile or remote energy applications [4].

Recent advancements in $S\text{-}CO_2$ power cycle modeling and optimization have increasingly adopted multi-objective evaluation methods to resolve the inherent trade-offs among various performance dimensions such as efficiency, compactness, and cost, aiming for optimal system-wide performance rather than focusing on the traditional single-objective pursuit of maximum thermal efficiency. For instance, Bakırcıoğlu et al. [5] proposed a solar-driven multigeneration system integrating an $S\text{-}CO_2$ recompression Brayton cycle, a transcritical CO_2 Rankine cycle, and a reverse-osmosis desalination unit, optimized using Non-dominated Sorting Genetic Algorithm II (NSGA-II) and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to maximize exergy efficiency and freshwater production while minimizing compressor work. Wang et al. [6] applied NSGA-II and TOPSIS to optimize a transcritical CO_2 cycle for marine engine exhaust heat recovery,

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Nomenclature

General

A	area (m^2)
C_i	relative closeness
D_i	distance
d	diameter (cm)
d_s	hydraulic diameter of the semi-circular channel (cm)
f	friction factor or volume fraction
G	mass flux ($kg/m^2/s$)
H	height (m)
HD	height-to-diameter ratio
HV	hypervolume
h	specific enthalpy (kJ/kg) or convective heat transfer coefficient($kW/m^2/K$)
I_j	amount of information
k_{eff}	effective multiplication factor
L	length (m)
M	mass (t) or decision matrix
$\dot{m}$	mass flow rate (kg/s)
m	multiplier
N	number of channels
Nu	Nusselt number
Pr	Prandtl number
p	pressure (MPa)
pr	pressure drop factor
p_c	crossover probability
p_m	mutation probability
$\dot{Q}$	heat power (kW)
q	power density (kW/m^3)
R	radius (cm) or resistance ($K \bullet m^3/kW$)
R^2	coefficient of determination
Re	Reynolds number
r	radius (cm)
S	strength (Pa)
s	spacing (cm)
s_j	standard deviation
T	temperature (K or ℃)
t	longitudinal pitch (cm)
U	heat transfer coefficient ($kW/m^2/K$)
UA	thermal conductance (kW/K)
u	flow velocity (m/s)
$\dot{W}$	power (kW)
w	specific power (kJ/kg)
w_j	weight
X	sample matrix
x	split ratio
x_{ij}	X matrix element
x_{ij}^*	X matrix element after normalized
Y	pareto front matrix
y_{ij}	Y matrix element
y_{ij}^*	Y matrix element after normalized

Greek symbols

η	efficiency
φ	UA distribution ratio
ε	fuel enrichment
λ_f	thermal conductivity of fluid (W/m / K)
λ_s	Thermal conductivity of solid (W/m / K)
ρ_f	density of fluid (kg/m^3)
ρ_s	density of solid (kg/m^3)
ρ_{ij}	correlation coefficient
μ	dynamic viscosity (Pa·s)
δ	thickness (cm)

Subscripts

1,2…14	state points
±	positive ideal or negative ideal
c	cross section
cal	calculated value
cool	coolant
clad	cladding
h/c	hot/cold side
i/o	inter/outer
in	input
lg	logarithmic mean
net	net power
pred	predicted value
pv	pressure vessel
refl	reflector
s	isentropic

Abbreviations

AHP	analytic hierarchy process
BOL	beginning of life
CRITIC	criteria importance through intercriteria correlation
EOL	end of life
HEX	heat exchanger
HTR	high-temperature recuperator
KPIs	key performance indicators
LCOE	levelized cost of electricity
LINMAP	linear programming technique for multidimensional analysis of preference
LMTD	logarithmic mean temperature difference
LTR	low-temperature recuperator
MC	main compressor
MCDM	multi-criteria decision-making
MLP	multilayer perceptron
MSE	mean squared error
NSGA-II	non-dominated sorting genetic algorithm II
PCHE	printed circuit heat exchanger
RC	re-compressor
$S\text{-}CO_2$	supercritical carbon dioxide
SMRs	small modular reactors
T	turbine
TID	tube-in-duct
TOPSIS	technique for order preference by similarity to ideal solution

incorporating thermodynamic analysis, sensitivity evaluation, and pinch point assessment, with the objectives of maximizing power output and minimizing both the heat exchanger area per unit power and levelized cost of electricity (LCOE). The multi-objective optimization results demonstrated improved thermodynamic performance, economic efficiency, and compactness of the cascaded CO_2 cycle power system. Li et al. [7] examined the thermodynamic and economic performance of an $S\text{-}CO_2$ power module coupled to a lead-cooled fast reactor, identifying the recompression cycle as the preferred configuration for its superior thermo-economic performance and spatial compactness. Zhao et al. [8] employed a multi-objective particle swarm optimization algorithm combined with the best compromise optimization scheme through the

TOPSIS decision-making method to optimize the marine nuclear power secondary loop, achieving reductions in weight and volume by 10.57 % and 13.68 %, respectively, while improving efficiency by 4.26 %. Du et al. developed a marine S-CO_2 nuclear system based on a small lead-cooled fast reactor and conducted a series of optimization studies. In their initial work [9], they focused on the size optimization and thermo-economic assessment of the heat exchangers. Building on this foundation, they performed a multi-objective optimization of the system's thermal efficiency, volume, specific energy and economic cost [10]. Most recently, they advanced the methodology by employing deep neural networks and data-mining techniques to achieve the optimal design of the recompression cycle [11]. Gao et al. [12] explored the influence of the Printed Circuit Heat Exchanger (PCHE) geometry and flow channel layout on system thermal efficiency and LCOE for S-CO_2 small modular reactors (SMRs), demonstrating that system-level performance is strongly constrained by flow-channel type. Jin et al. [13] accelerated the multi-objective optimization of a recompression S-CO_2 waste-heat recovery cycle using artificial neural networks, evaluating net power output, exergy efficiency, and ecological function via TOPSIS, Linear Programming Technique for Multidimensional Analysis of Preference (LINMAP), and entropy-based weighting.

A comparative summary of representative S-CO_2 optimization studies is presented in Table 1. Despite notable progress, research gaps remain for direct-cycle S-CO_2 nuclear systems, particularly regarding the strong coupling between reactor physics (neutron reactivity and core heat output) and thermodynamic cycle performance. Most existing works treat the nuclear reactor and power conversion subsystem independently, limiting insight into the interdependent design trade-offs among neutronic safety margins, heat-source conditions, total system mass, and efficiency.

Furthermore, many current multi-objective optimization frameworks still rely on subjective weighting or independent objective treatment, neglecting the intrinsic correlations among objectives and thereby reducing decision-making objectivity and representativeness. This highlights the need for a unified, physics-constrained, and data-driven optimization–decision framework that can systematically balance performance, mass, and safety across the entire reactor–cycle coupled system.

Therefore, this study develops an integrated modeling, surrogate-assisted multi-objective optimization, and objective decision-making framework for a direct S-CO_2 Brayton cycle nuclear energy system. The proposed approach explicitly couples the reactor core neutronic and thermal–hydraulic models with the power conversion system, and introduces a Criteria Importance Through Intercriteria Correlation (CRITIC)–TOPSIS-based decision strategy to enhance objectivity, reduce subjectivity, and ensure the comprehensive representativeness of the optimization outcomes.

The remainder of this paper is organized as follows. Section 2 details the system modeling methods, including the thermodynamic cycle model, the heat exchanger mass model, the reactor mass model, and reactor-cycle coupling model, forming the foundation for performance analysis, and multi-objective optimization and decision-making method. Section 3 presents validation results, including validation of the thermodynamic model, reactor design optimization and performance evaluation, sensitivity analysis of key parameters, and the outcomes of multi-objective optimization and decision-making. Finally, Section 4 concludes the study and discusses implications for future research.

Table 1
Comparative analysis of representative optimization objectives and methods for S-CO_2 cycle studies.

Research	Optimization object	Optimization objectives	Decision method	Weighting method
Bakırcıoğlu [5]	Solar-powered multigeneration plant (S-CO_2 Brayton cycle + transcritical CO_2 Rankine cycle + Reverse Osmosis desalination)	Compressor work, Fresh water rate, Exergy efficiency	TOPSIS	/
Wang [6]	Transcritical CO_2 cycle for marine engine exhaust heat recovery	Net output power, Heat exchanger area per unit power, LCOE	TOPSIS	/
Li [7]	S-CO_2 lead-cooled fast reactor power cycle	Thermal efficiency, Electricity production costs	/	/
Zhao [8]	Marine nuclear power secondary loop system	Mass, Volume, Thermal efficiency	TOPSIS	/
Du [10]	Marine lead-cooled fast reactor S-CO_2 nuclear system	Thermal efficiency, Volume, Specific energy, Economic cost	LINMAP	/
Gao [12]	S-CO_2 Brayton cycle for SMRs	Thermal Efficiency, LCOE	TOPSIS	/
Jin [13]	S-CO_2 waste heat recovery re-compression power cycle	Net output power, Exergy efficiency, Thermal efficiency, Exergy output	TOPSIS /LINMAP /Shannon Entropy	–/-/ Entropy weighting
This study	Integrated S-CO_2 nuclear system (cycle + reactor)	Thermal efficiency, Specific work, System mass	TOPSIS	Objective, data-driven weighting

2. Methods

The S-CO_2 recompression Brayton cycle is adopted as the power conversion system for the S-CO_2-cooled nuclear reactor, as illustrated in Fig. 1. To enable a comprehensive performance evaluation and optimization of the system, key performance indicators (KPIs) are identified based on a review of state-of-the-art studies across nuclear, solar-thermal, and marine propulsion applications. These KPIs are classified into two main categories: thermodynamic performance metrics (e.g., thermal efficiency and specific power output) and lightweight performance metrics (e.g., reactor mass and heat exchanger mass).

This section establishes the modeling and optimization framework used for the integrated S-CO_2 nuclear energy system. The following subsequent describe in detail the thermodynamic cycle model, the reactor mass model, and the heat exchanger mass model, which collectively form the analytical foundation for the subsequent optimization study.

The system mass plays a pivotal role in determining the transportability, ease of installation, and operational stability. Prior studies

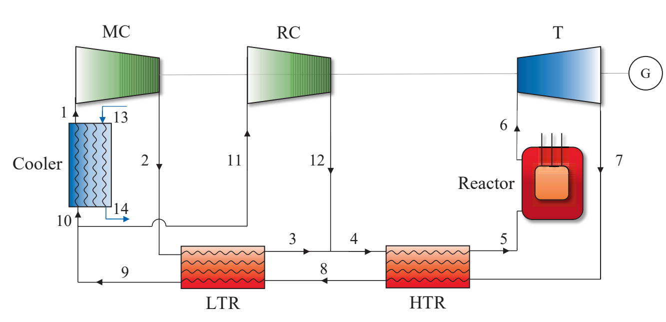


Fig. 1. Schematic layout of the S-CO_2 nuclear energy system.

[14] indicate that turbomachinery components (compressors and turbines) contribute relatively little to total system mass, accounting for only a few percent of total system mass compared to other components. Therefore, the lightweight assessment focuses primarily on heat exchangers and the nuclear reactor. Heat exchangers, as high-mass elements, significantly influence both system compactness and thermal efficiency. The reactor mass, on the other hand, not only dictates the overall system weight but also affects critical neutronic parameters, such as fuel volume fraction and reactivity margin. Finally, a multi-objective optimization and objective decision-making method is implemented to achieve an optimal balance among conflicting design objectives—ensuring a lightweight yet thermodynamically efficient and safe system configuration.

2.1. Thermodynamic cycle model

To accurately establish the thermodynamic model of the S-CO_2 power cycle, key component sub-models are developed with reference to well-established modeling libraries for power cycle systems. These models employ rational node segmentation and appropriate component simplification strategies. The primary input parameters adopted in this study are summarized in Table 2.

For the compressor and turbine models, the isentropic efficiency is employed to characterize internal irreversibility. The actual power output ($\dot{W}_{\mathrm{T}}$) or consumption ($\dot{W}_{\mathrm{MC}}$ and $\dot{W}_{\mathrm{RC}}$) of each turbomachine is expressed as:

$$\dot{W}_{\mathrm{T}} = \dot{m}_{\mathrm{CO_2}}(h_{6,s} - h_7)\eta_{\mathrm{T}}, \tag{1}$$

$$\dot{W}_{\mathrm{MC}} = (1-x)\dot{m}_{\mathrm{CO_2}}\frac{(h_2 - h_{1,s})}{\eta_{\mathrm{MC}}}, \tag{2}$$

$$\dot{W}_{\mathrm{RC}} = x\dot{m}_{\mathrm{CO_2}}\frac{(h_{12} - h_{11,s})}{\eta_{\mathrm{RC}}}, \tag{3}$$

where $\dot{m}_{\mathrm{CO_2}}$ is the total mass flow rate, h denotes the specific enthalpy, and the subscripts correspond to node numbers shown in Fig. 1. The subscript s indicates an isentropic process and the mass flow split ratio x defines the fraction of S-CO_2 directed to the re-compressor.

The heat transfer in LTR, HTR and cooler is computed using the Logarithmic Mean Temperature Difference (LMTD) method:

$$\dot{Q}_{\mathrm{LTR}} = \dot{m}_{\mathrm{CO_2}}(h_3 - h_2) = \dot{m}_{\mathrm{CO_2}}(h_8 - h_9) = \varphi_{\mathrm{LTR}} UA_{\mathrm{Total}}\frac{T_9 - T_2 - T_8 + T_3}{ln\frac{T_9 - T_2}{T_8 - T_3}}, \tag{4}$$

Table 2
Basic input parameters for the thermodynamic model.

Parameters	Value
Net power output ($\dot{W}_{\mathrm{net}}$) [15]	10 MW
Main Compressor (MC) inlet pressure (p_1)	7.8 MPa
MC inlet temperature (T_1) [16]	305.15 K
Turbine inlet pressure (p_6)	25 MPa
Turbine inlet temperature (T_6) [16]	823.15 K
Cooling water inlet pressure (p_{13}) [17]	0.5 MPa
Cooling water inlet temperature (T_{13}) [17]	293.15 K
Total heat conductance (UA_{Total}) [13]	2500 kW/K
UA distribution ratio of Low-Temperature Recuperator (LTR) (φ_{LTR}) [13]	0.3
UA distribution ratio of High-Temperature Recuperator (HTR) (φ_{HTR}) [13]	0.3
MC isentropic efficiency (η_{MC}) [18]	0.89
Re-Compressor (RC) isentropic efficiency (η_{RC}) [18]	0.89
Turbine isentropic efficiency (η_{T}) [18]	0.93
Mass flow split ratio (x)	0.3
Heat exchanger initial pressure factor (pr)	0.98

$$\dot{Q}_{\mathrm{HTR}} = \dot{m}_{\mathrm{CO_2}}(h_7 - h_8) = \dot{m}_{\mathrm{CO_2}}(h_5 - h_4) = \varphi_{\mathrm{HTR}} UA_{\mathrm{Total}}\frac{T_8 - T_4 - T_7 + T_5}{ln\frac{T_8 - T_4}{T_7 - T_5}}, \tag{5}$$

$$\begin{aligned}\dot{Q}_{\mathrm{Cooler}} &= \dot{m}_{\mathrm{CO_2}}(h_{10} - h_1) = \dot{m}_{\mathrm{Cooler}}(h_{14} - h_{13}) \\ &= (1 - \varphi_{\mathrm{LTR}} - \varphi_{\mathrm{HTR}})UA_{\mathrm{Total}}\frac{T_1 - T_{13} - T_{10} + T_{14}}{ln\dfrac{T_1 - T_{13}}{T_{10} - T_{14}}}.\end{aligned} \tag{6}$$

Here, $\dot{Q}$ represents the heat transfer rate, and UA_{Total} denotes the total heat conductance. φ_{LTR} and φ_{HTR} are the allocation ratios of total heat conductance, defined as the proportions of UA_{Total} assigned to LTR and HTR, respectively; and T_i is the temperature at the corresponding node i.

The nuclear reactor heat input to the power cycle is given by:

$$\dot{Q}_{\mathrm{Reactor}} = \dot{m}_{\mathrm{CO_2}}(h_6 - h_5). \tag{7}$$

The model is solved by combining mass and energy conservation equations with component performance relations, forming a nonlinear system. These equations are iteratively solved using the global Newton [19] iterative method. Thermophysical properties of CO_2 and water are computed using CoolProp [20], an open-source library that provides high-fidelity equations of state.

Two performance indicators are used to evaluate the system:

$$\eta_{\mathrm{cycle}} = \frac{\dot{W}_{\mathrm{net}}}{\dot{Q}_{\mathrm{in}}} = \frac{\dot{W}_{\mathrm{T}} - \dot{W}_{\mathrm{MC}} - \dot{W}_{\mathrm{RC}}}{\dot{Q}_{\mathrm{Reactor}}}, \tag{8}$$

$$w = \frac{\dot{W}_{\mathrm{net}}}{\dot{m}_{\mathrm{CO_2}}} = \frac{\dot{W}_{\mathrm{T}} - \dot{W}_{\mathrm{MC}} - \dot{W}_{\mathrm{RC}}}{\dot{m}_{\mathrm{CO_2}}}. \tag{9}$$

The parameter η_{cycle} characterizes the overall conversion efficiency of reactor heat into useful work, while w denotes the specific power output per unit mass flow. Higher efficiency indicates more efficient utilization of reactor heat, whereas a higher specific work enhances system compactness and power density, which are particularly critical under mass- and volume-constrained conditions such as marine or space-based nuclear systems.

2.2. Heat exchanger mass model

In this study, PCHEs with semi-circular straight flow channels are employed for the LTR, HTR and cooler. The cross-sectional configuration is shown in Fig. 2, where d_c represents the diameter of the semi-circular channel, t_c is the longitudinal pitch (distance between adjacent hot and cold fluid channels), and s_c is the transverse pitch (spacing between the neighboring semi-circular channels). The PCHEs are constructed from Inconel 617, with density $\rho_s = 8360\ \mathrm{kg/m^3}$ and thermal conductivity $\lambda_s = 21\ \mathrm{W/(m \bullet K)}$ [21].

The total number of flow channel is determined by the mass flux G,

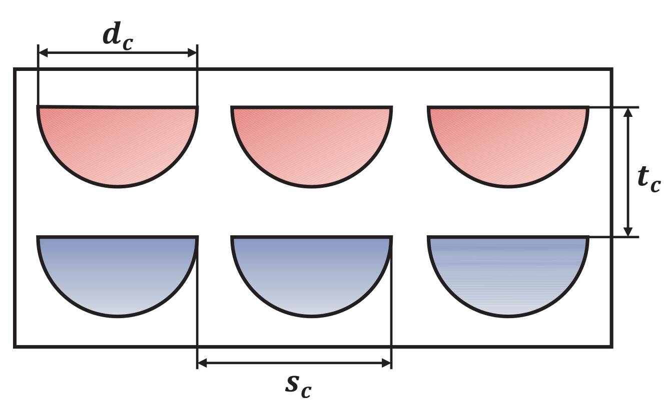


Fig. 2. Cross-sectional schematic of the PCHE channel layout.

which dictates the velocity of the working fluid through each channel. The number of channels is calculated as:

$$N = \frac{\dot{m}}{A_c \bullet G} = \frac{8\dot{m}}{\pi d_c^2 \bullet G}, \tag{10}$$

where $\dot{m}$ represents the mass flow rate and A_c is the cross-sectional area of one channel.

The convective heat transfer coefficient inside the semi-circular channels is calculated using the Gnielinski [22] correlation:

$$Nu = \begin{cases} \dfrac{(f/8)(Re-1000)Pr}{1+12.7(Pr^{\frac{2}{3}}-1)\sqrt{\dfrac{f}{8}}}, Re \geq 5000, \\ 4.089 + \dfrac{Nu_{Re=5000}-4.089}{5000-2300}(Re-2300), 2300 \leq Re < 5000, \\ 4.089, Re < 2300, \end{cases} \tag{11}$$

where Nu represents the Nusselt number; $Re = G \bullet d_s/\mu$ is the Reynolds number, d_s is the hydraulic diameter of the semi-circular channel, μ is the dynamic viscosity, and Pr is the Prandtl number. The friction factor f is given by:

$$f = \begin{cases} \left(\dfrac{1}{1.8log(Re)-1.5}\right)^2, Re \geq 2300, \\ 64/Re, Re < 2300. \end{cases} \tag{12}$$

Once Nu is determined, the convective heat transfer coefficient $h_{h/c}$ and the overall heat transfer coefficient U are calculated by:

$$h_{h/c} = \frac{Nu \bullet \lambda_f}{d_s}, \tag{13}$$

$$U = \frac{1}{\frac{1}{h_h} + \frac{\lambda_s}{t_c} + \frac{1}{h_c}}, \tag{14}$$

where subscripts h and c denote the hot and cold sides, respectively. λ_f and λ_s represent the thermal conductivities of the working fluid and solid material. Given the overall heat exchanger UA value from the S-CO_2 power cycle model, the heat exchanger area A and channel length L can be determined by:

$$A = \frac{\dot{Q}}{U \cdot \Delta T_{lg}}, \tag{15}$$

$$L = \frac{A}{\left(\frac{\pi}{2}+1\right)d_c \bullet N}, \tag{16}$$

where $\dot{Q}$ represents the total heat transfer rate and ΔT_{lg} is the logarithmic mean temperature difference.

For the pressure loss in the heat exchanger, only frictional pressure losses inside the channels are considered, neglecting local inlet/outlet effects. The pressure drop Δp_{cal} is given by:

$$\Delta p_{cal} = \frac{fL\rho_f u^2}{2d_s}, \tag{17}$$

where ρ_f and u denote the fluid density and average velocity, respectively. An iterative procedure is adopted to reconcile this calculated pressure drop with the input specification until convergence is achieved. Once channel geometry and layout are finalized, the mass of the PCHEs is determined as:

$$M_{HEX} = \left(1 - \frac{\pi d_c^2}{8 s_c t_c}\right)\rho_s L N s_c t_c, \tag{18}$$

which accounts for the solid metal fraction of the PCHE structure. This formulation enables precise mass estimation for each exchanger under varying geometric and operating conditions, thus supporting the lightweight optimization objectives of the coupled reactor–cycle system.

2.3. Reactor mass model

The reactor model considered in this study consists of three primary structural components: the core, the reflector, and the pressure vessel, as illustrated in Fig. 3. The selected materials and key design parameters are summarized in Tables 3 and 4, where the fuel enrichment listed in Table 4 is set according to Ref. [23], while the material selection in Table 3 and the remaining design parameters in Table 4 are primarily based on representative data from Ref. [24].

The core region adopts a hexagonal Tube-in-Duct (TID) fuel element configuration, differing from the traditional rod-type fuel with inter-rod coolant channels. Compared with conventional rod-type designs, the TID configuration achieves a higher fuel volume fraction, which reduces the coolant's volume fraction and moderation effect, minimizes the positive reactivity contribution from coolant voids and improves the overall core reactivity margin [25].

For mass estimation and parametric optimization, three key design aspects are selected: core size, fuel mass, and reflector thickness. Specifically, the core size is characterized by the core radius R_{core}, ranging from 25 cm to 95 cm. The fuel mass is determined jointly by the fuel volume fraction f_{fuel} (0.2–0.9) and the core radius R_{core}. Since reflector thickness δ_{refl} inherently depends on the core radius, it is represented by a relative thickness multiplier m_{refl} (0.5–0.9). The relationships among these variables are formulated in Appendix A.1.

To improve computational efficiency during optimization, the reactor core materials are treated using a homogenization approach, in which fuel, cladding, and coolant regions are represented by volume-weighted averaged properties. This approximation has been validated to accurately capture global neutronic behavior while substantially reducing the computational cost, as demonstrated in Appendix A.1.1.

2.3.1. Reactivity surrogate model

The reactivity surrogate model is developed to establish a quantitative mapping between reactor design parameters and the effective neutron multiplication factor k_{eff}, as defined in Eq. (19):

$$k_{eff} = f\left(R_{core}, f_{fuel}, m_{refl}\right). \tag{19}$$

This surrogate model replaces computationally expensive Monte Carlo neutronic calculations with a machine-learning-based approximation that captures the underlying nonlinear dependencies among design variables. The model enables rapid reactivity evaluations within the multi-objective optimization framework.

A fully connected Multilayer Perceptron (MLP) neural network [26]

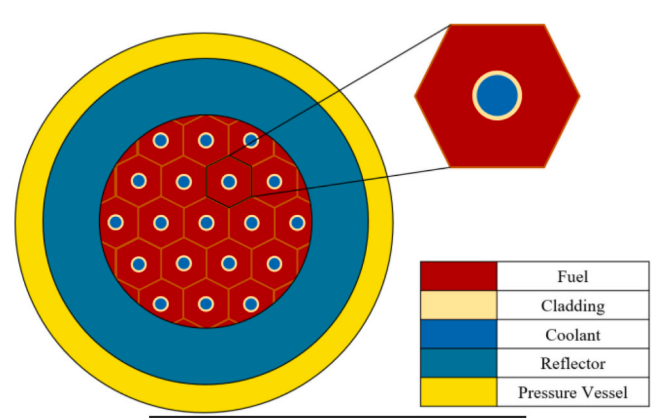


Fig. 3. Diagram of the reactor.

Table 3
Material composition of the S-CO_2 reactor.

Component	Material
Fuel	UO_2
Coolant	S-CO_2
Cladding	Inconel-718
Reflector	BeO
Pressure vessel	SS304

Table 4
Basic parameter settings of the S-CO_2 reactor.

Parameter	Symbol	Unit	Value
Fuel enrichment	ε	%	19.85
Core height-to-diameter ratio	HD_{core}	–	1.5
Coolant channel radius	r_{cool}	cm	0.5
Cladding thickness	δ_{clad}	cm	0.031
Fuel density	ρ_{fuel}	g/cm^3	10.97
Reflector density	ρ_{refl}	g/cm^3	3.01
Cladding density	ρ_{clad}	g/cm^3	8.20
Pressure vessel density	ρ_{pv}	g/cm^3	8.05
Reactor lifetime	–	a	10

is adopted to construct the surrogate model, as illustrated in Fig. 4. The MLP consists of one input layer, two hidden layers, and one output layer. The input layer receives the normalized reactor design variables [R_{core}, f_{fuel}, m_{refl}]. The hidden layers employ nonlinear activation functions to extract complex correlations and nonlinearities, while the output layer produces the predicted value of k_{eff}.

To train the MLP, a large dataset comprising 8000 reactor configurations is generated using the OpenMC Monte Carlo code [27]. The dataset is divided into training (70 %), validation (15 %), and testing (15 %) subsets: The training set is used to optimize network parameters; the validation set is employed to monitor convergence and prevent overfitting; and the testing set is used to assess predictive generalization.

The neural network is trained using the Adam optimizer with an initial learning rate of 0.001, which decays by a factor of 0.9 every 500 steps to ensure convergence stability. The training objective minimizes the Mean Squared Error (MSE) between predicted and reference k_{eff} values. Model accuracy is evaluated using both the MSE and the coefficient of determination (R^2), defined respectively as:

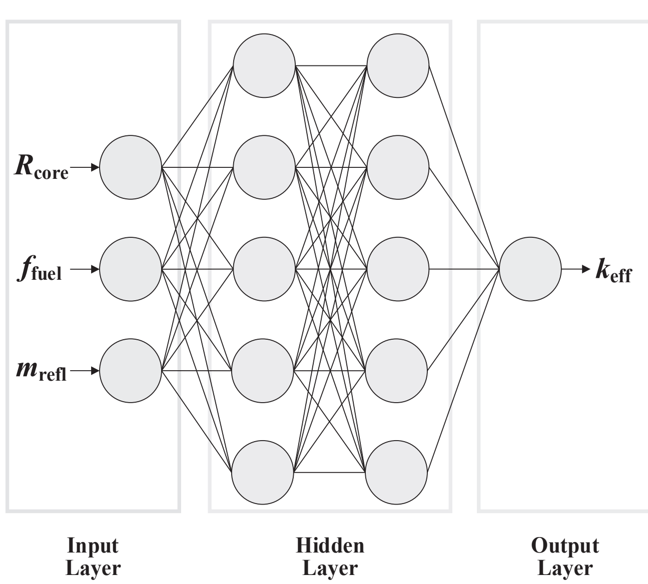


Fig. 4. Schematic architecture of the neural network surrogate model.

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(k_{\mathrm{eff},i}^{\mathrm{pred}} - k_{\mathrm{eff},i}^{\mathrm{MC}}\right)^2, \tag{20}$$

$$R^2 = 1 - \frac{\sum_i\left(k_{\mathrm{eff},i}^{\mathrm{pred}} - k_{\mathrm{eff},i}^{\mathrm{MC}}\right)^2}{\sum_i\left(\overline{k}_{\mathrm{eff}}^{\mathrm{MC}} - k_{\mathrm{eff},i}^{\mathrm{MC}}\right)^2}. \tag{21}$$

Here, $k_{\mathrm{eff},i}^{\mathrm{pred}}$ and $k_{\mathrm{eff},i}^{\mathrm{MC}}$ denote the k_{eff} value for the i-th sample predicted by the neural network and obtained from OpenMC simulations, respectively. $\overline{k}_{\mathrm{eff}}^{\mathrm{MC}}$ represents the mean value of all reference Monte Carlo k_{eff} results.

Meanwhile, an early stopping mechanism is incorporated to prevent overfitting by continuously monitoring the validation loss; training is terminated when no improvement is observed over a predefined number of consecutive iterations. This strategy improves the generalization capability while maintaining numerical stability.

2.3.2. *Mass optimization model under critical reactivity constraint*

(1) Reactivity margin assessment

Reactor reactivity serves as a critical physical constraint in nuclear system design. During the operational lifetime, continuous fuel depletion leads to a gradual decrease in the effective neutron multiplication factor k_{eff}. To ensure sustained criticality, the reactor must maintain k_{eff} > 1 at the End of Life (EOL). This requirement necessitates that sufficient reactivity margin k_{margin} should be reserved at the Beginning of Life (BOL).

Moreover, the thermal power output of the reactor must satisfy the power coupling condition with the S-CO_2 Brayton cycle. Under a fixed net power output of 10 MW, the reactor's total thermal power must provide adequate margin for system efficiency and power matching. In this study, two representative design cases—20 MW and 30 MW reactor core power—are investigated to capture different operational scenarios.

A series of fuel depletion simulations are carried out using the OpenMC code to systematically evaluate how variations in the core design parameters $\left(R_{core}, f_{fuel}, m_{refl}\right)$ affect reactivity depletion behavior. The corresponding BOL reactivity margin is quantified and subsequently used as a physics-based constraint in the reactor mass optimization process.

(2) Reactor size matching under reactivity constraint

Using the trained surrogate model for k_{eff} prediction, a criticality matching optimization framework is developed to identify the optimal reactor geometry that satisfies the reactivity margin for varying fuel fractions. The design space defined by f_{fuel} (fuel volume fraction) and m_{refl} (reflector thickness multiplier) is discretized into an $m \times n$ grid. For each pair ($f_{fuel,i}$ with $i = 1, 2, \cdots, m$, $m_{refl,j}$ with $j = 1, 2, \cdots, n$), an optimization objective function J_1 is formulated to minimize the deviation between the predicted k_{eff} and the margin value k_{margin}, as defined by:

$$\min J_1 = \min_{R_{core}}\left(k_{eff}\left(R_{core}, f_{fuel,i}, m_{refl,j}\right) - k_{margin}\right)^2. \tag{22}$$

This optimization problem is solved using the SciPy [28] optimization library, enabling efficient convergence to the optimal core radius R_{core} that satisfies the reactivity constraint.

(3) Reflector thickness adjustment and mass optimization

Once the optimal R_{core} is obtained, a second-stage optimization is performed to minimize the total reactor mass $M_{Reactor}$ while maintaining the required reactivity margin k_{margin} constraint. This step adjusts the reflector thickness multiplier m_{refl}, with the objective function defined as:

$$\min J_2 = \min_{m_{refl}} M_{Reactor}\left(R_{core}, f_{fuel,i}, m_{refl}\right). \tag{23}$$

Here, the total reactor mass $M_{Reactor}$ comprises the masses of the core, reflector, and pressure vessel, all of which are parameterized in terms of geometry and material density defined from Eq. (24) to (27):

$$M_{\text{Reactor}} = M_{\text{core}} + M_{\text{refl}} + M_{\text{pv}}, \tag{24}$$

$$M_{\text{core}} = \sum \rho_i A_i H_{\text{core}}, i = [\text{fuel}, \text{cool}, \text{clad}], \tag{25}$$

$$M_{\text{refl}} = \rho_{\text{refl}} \bullet A_{\text{refl}} \bullet (H_{\text{core}} + 2\delta_{\text{refl}}), \tag{26}$$

$$M_{\text{pv}} = \rho_{\text{pv}} \bullet A_{\text{pv}} \bullet (H_{\text{core}} + 2\delta_{\text{refl}} + 2\delta_{\text{pv}}). \tag{27}$$

Here, M and A represent the mass and cross-sectional area of the various components of each reactor component, respectively. This optimization yields, for each fuel fraction $f_{\text{fuel},i}$, the corresponding optimal reflector thickness and corresponding reactor configuration. This ensures an optimal balance between neutronic performance, structural compactness, and mass minimization, thereby providing a physically consistent and computationally efficient foundation for the integrated reactor–-cycle optimization process.

2.4. Reactor-cycle coupled thermal power matching model

To ensure consistency between the reactor's neutronic–thermal performance and the power conversion requirements of the S-CO_2 Brayton cycle, a coupled thermal power matching model is established. Building upon the reactivity-matching and reactor-mass optimization framework introduced earlier, this model quantitatively links reactor design variables with the thermal–hydraulic characteristics and the heat demand of the power cycle, ensuring that the reactor core can supply the required thermal power while minimizing structural mass.

As illustrated in Fig. 5, the model integrates the reactor's geometric parameters, coolant distribution, and system power requirements into a unified optimization loop. Within this framework, the coolant velocity in each core channel is determined jointly by the total mass flow rate entering the reactor and the coolant volume fraction within the core. The former is constrained by the thermal power requirement of the S-CO_2 power cycle, whereas the latter depends on the reactor geometry and fuel loading, which govern the core's power density and hydraulic resistance.

The S-CO_2 cycle provides the key boundary conditions for coupling, including the inlet/outlet temperature (T_5/T_6), pressure (p_5/p_6), mass flow rate ($\dot{m}_{\text{cool}}$), and the required reactor thermal power ($\dot{Q}_{\text{Reactor}}$). Given a prescribed fuel volume fraction f_{fuel}, the corresponding reactor geometric parameters are derived from the reactivity-matching results. The computed reactor core thermal power $\dot{Q}_{\text{cal}}$ is evaluated using thermal–hydraulic model described in Ref. [29], with the formulation provided in Appendix A.1.4.

The thermal power matching process then optimizes f_{fuel} to minimize

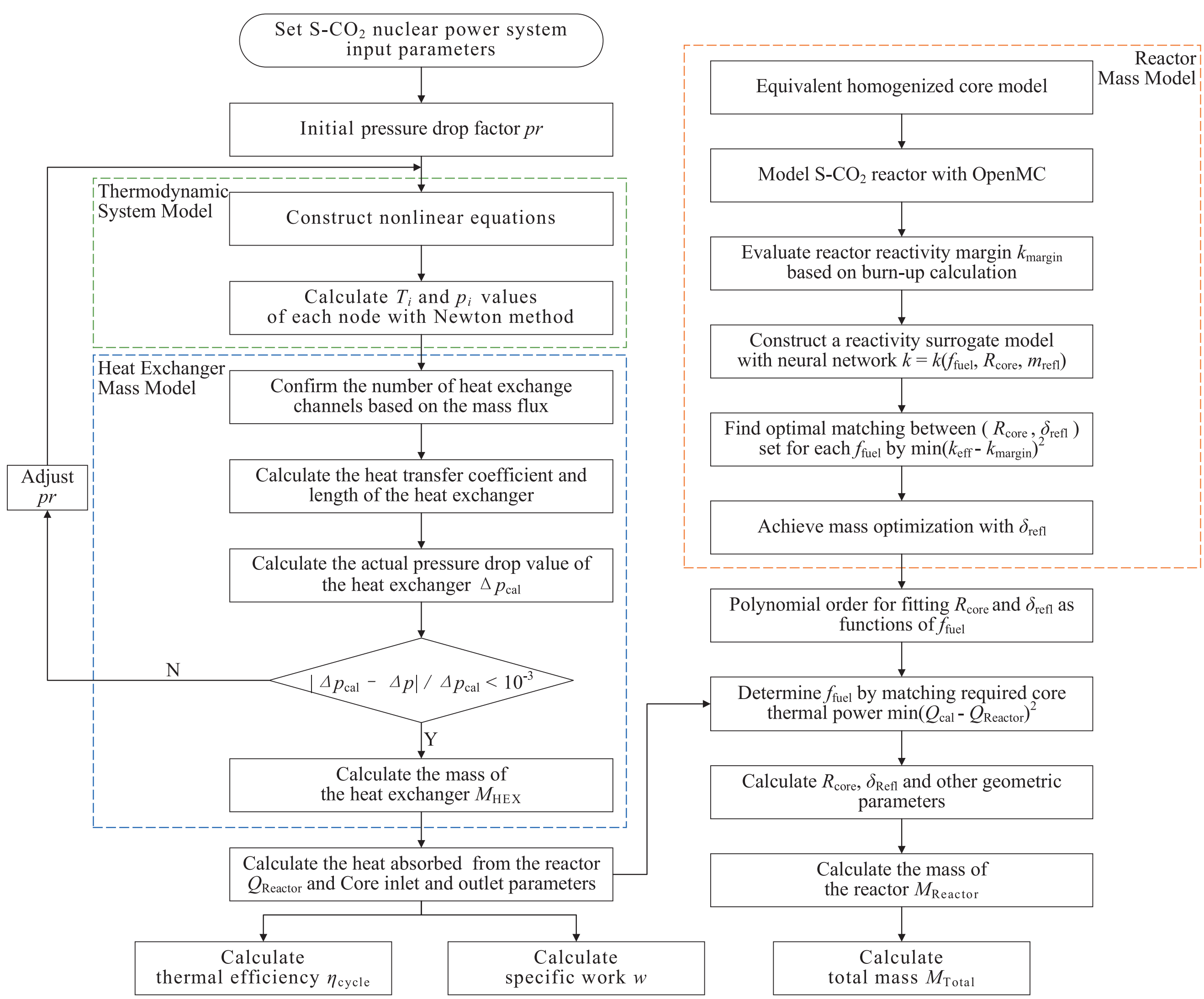


Fig. 5. Flowchart of reactor-cycle coupling thermal power matching model.

the deviation between the calculated and required thermal power, as defined by the objective function:

$$\min f = \min\left(\dot{Q}_{\text{cal}} - \dot{Q}_{\text{Reactor}}\right)^2. \tag{28}$$

Once the matching condition is achieved, a pressure-drop validation is performed. The calculated pressure drop Δp_{cal} across the reactor core is compared against the allowable pressure difference (p_4 −p_5) from the S-CO_2 cycle model. If the constraint is violated, the design variables—namely f_{fuel} and the number of coolant channels—are iteratively updated until both thermal and hydraulic criteria are simultaneously satisfied. The final matched reactor configuration thus guarantees accurate heat-source delivery and system-level compatibility for subsequent optimization stages.

2.5. Multi-objective optimization and decision-making method

In complex engineering systems, multiple objectives often conflict with one another—such as maximizing efficiency, minimizing mass, and enhancing specific power—making it impossible to identify a single globally optimal solution. Instead, a family of non-dominated solutions, collectively known as the Pareto front, is obtained, where improvement in one objective necessarily results in degradation of at least one other. Each Pareto-optimal point thus represents a distinct design trade-off reflecting different engineering preferences or operational priorities.

However, for practical system design, a unique and representative configuration must often be selected from the Pareto front. To achieve this, multi-criteria decision-making (MCDM) techniques are employed to evaluate each Pareto solution based on a set of weighted performance indices. By quantifying the relative importance of competing objectives, MCDM methods transform the Pareto front into a single compromise solution, effectively bridging the gap between mathematical optimization and engineering implementation.

2.5.1. Optimization method

To handle the multi-objective optimization of the integrated S-CO_2 nuclear energy system, this study employs the NSGA-II [30], an established evolutionary algorithm known for its robustness in exploring complex, non-convex search spaces. NSGA-II combines fast non-dominated sorting, elitism, and crowding distance mechanisms to ensure both convergence toward the true Pareto front and a well-distributed set of trade-off solutions.

The algorithmic workflow used in this study is illustrated in Fig. 6. Each iteration evaluates the thermodynamic and structural performance of candidate designs using the coupled reactor–cycle model and the surrogate-assisted submodels described in previous sections.

The optimization simultaneously considers three conflicting objectives: maximizing the thermal efficiency η_{cycle}, maximizing the specific power w, and minimizing the total system mass M_{Total}. The design variables and their respective search ranges are summarized in Table 5.

To balance convergence speed, solution diversity, and computational cost, the NSGA-II parameters are configured with a population size of 200, a maximum of 500 generations, and crossover and mutation probabilities of 0.95 and 0.01, respectively. Specifically, a population of 200 ensures adequate design diversity while maintaining manageable computational time for each iteration, considering the multi-physics evaluation cost. Meanwhile, the crossover probability (p_c = 0.95) promotes exploration and accelerating Pareto front growth in the early stages of optimization without harming late-stage refinement, whereas the mutation probability (p_m = 0.01) maintains population diversity and prevents premature convergence without introducing excessive randomness.

The convergence and spread of Pareto solutions are quantitatively assessed using the Hypervolume (HV) indicator, which measures the dominated volume in the objective space. A larger HV value implies better front convergence and diversity. In this study, HV values are computed at each generation to monitor algorithmic performance and ensure the quality of the final Pareto front.

2.5.2. Decision making method

To select the most appropriate configuration from the Pareto-optimal set, this study employs a hybrid decision-making method combining CRITIC method for objective weighting and TOPSIS method for alternative ranking, which involves constructing a decision matrix, computing objective weights, normalizing the data, and ranking solutions based on a composite score as shown in Fig. 7.

1) CRITIC method for objective weighting

The CRITIC [31] method determines the relative importance of each performance criterion by analyzing both its contrast intensity (via standard deviation) and conflict level (via inter-criterion correlation). Unlike traditional subjective weighting schemes such as the Analytic Hierarchy Process (AHP), CRITIC objectively quantifies each criterion's

Table 5
Range of optimized design variables.

Optimized variables	Range	Optimized variables	Range
p_1/MPa	7.5–9.5	x	0.1–0.5
p_6/MPa	15–25	φ_{LTR}, φ_{HTR}	0.1–0.5
UA_{Total}/(kW/K)	1500–3000	G_{LTR}, G_{HTR}, G_{Cooler}/[kg/(m^2·s)]	400–900

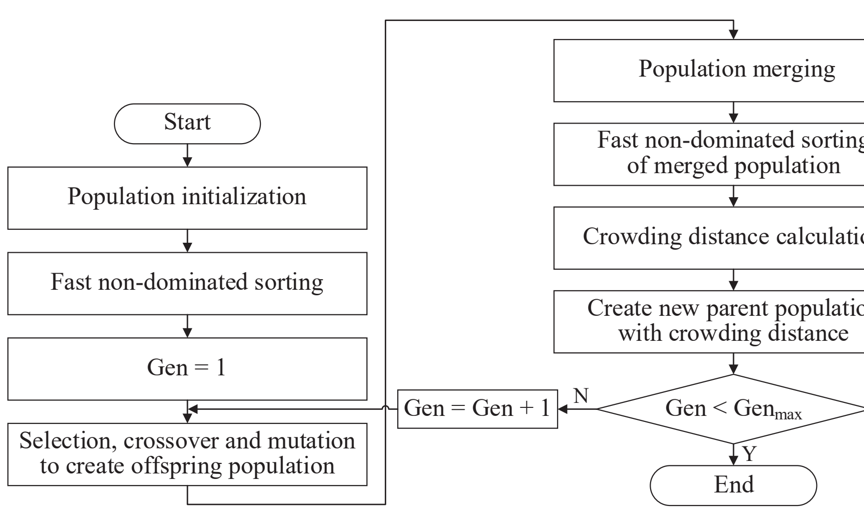


Fig. 6. Flowchart of the NSGA-II-based multi-objective optimization framework.

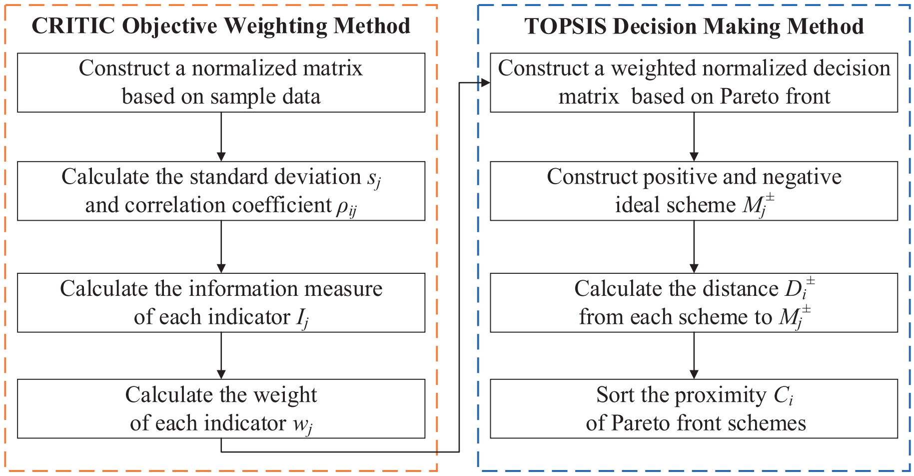


Fig. 7. Flowchart of CRITIC-TOPSIS method.

information contribution and mutual redundancy, ensuring an unbiased representation of its discriminative power.

A sample matrix $X = [x_{ij}]_{m\times n}$ is constructed within the range of the selected optimization variables, where x_{ij} represents the value of the j-th indicator for the i-th sample. Here, m is the number of sample data, and n is the number of objectives.

To eliminate the influence of unit and scale inconsistencies among the objectives, normalization is applied using Eq. (29).

$$x_{ij}^{*} = \begin{cases} \dfrac{x_{ij} - x_{j,\min}}{x_{j,\max} - x_{j,\min}}, for positive objective, \\ \dfrac{x_{j,\max} - x_{ij}}{x_{j,\max} - x_{j,\min}}, for negative objective, \end{cases} \quad i = 1, 2, \cdots m; j = 1, 2, \cdots n. \tag{29}$$

The amount of information I_j contained in each objective is computed as:

$$I_j = s_j \sum_{i=1}^{n} (1 - \rho_{ij}), j = 1, 2, \cdots n, \tag{30}$$

where s_j denotes the standard deviation and ρ_{ij} the correlation coefficient between objectives i and j. The objective weight w_j is then obtained as:

$$w_j = \frac{I_j}{\sum_{j=1}^{n} I_j}. \tag{31}$$

2) TOPSIS method for decision-making

Once the objective weights w_j are determined, the TOPSIS [32] method is applied to rank Pareto-optimal solutions with matrix $Y = [y_{ij}]_{m\times n}$ by evaluating their geometric proximity to an ideal solution (best for all objectives) and distance from a negative ideal solution (worst for all objectives). The normalized decision matrix is constructed as:

$$y_{ij}^{*} = \frac{y_{ij}}{\sqrt{\sum_{j=1}^{n} y_{ij}^2}}, \tag{32}$$

and the weighted normalized decision matrix M is computed as:

$$M = w_j \bullet Y^{*}. \tag{33}$$

The Euclidean distances from the positive and negative ideal points are then expressed as:

$$D_i^{\pm} = \sqrt{\sum_{j}^{n} (M - M_j^{\pm})^2}, \tag{34}$$

where the positive ideal solution M_j^{+} and negative ideal solution M_j^{-} are determined by:

$$M_j^{+} = [\max(M_{i1}), \max(M_{i2}), \min(M_{i3})], \tag{35}$$

$$M_j^{-} = [\min(M_{i1}), \min(M_{i2}), \max(M_{i3})]. \tag{36}$$

The relative closeness coefficient C_i, which measures each solution's proximity to the ideal configuration, is calculated as:

$$C_i = \max\left(\frac{D_i^{-}}{D_i^{+} + D_i^{-}}\right), \quad i = 1, 2, \cdots m. \tag{37}$$

Finally, all Pareto solutions are ranked in descending order of C_i, and the one with the highest value is selected as the TOPSIS-optimal configuration — representing the most balanced trade-off between system efficiency, specific power, and mass.

3. Results and discussion

This section presents and discusses the main findings of the study, including model evaluation, reactor mass model evaluation, optimization results, and decision analysis. The discussion highlights the performance trade-offs among efficiency, specific work, and system mass, as well as the robustness of the optimized configurations.

3.1. Thermodynamic model validation

To verify the accuracy and reliability of the developed thermodynamic model for the S-CO_2 recompression Brayton cycle, a validation study is conducted by comparing the predicted results with well-established reference data under equivalent operating conditions. As summarized in Table 6, the thermal efficiency predicted by the present model shows excellent agreement with both references, with relative deviations below 1.5 %. This confirms that the proposed model can

Table 6
Comparison of thermal efficiency from present thermodynamic model and literature.

Reference	Literature results	Present models	Relative errors
Dyreby [16]	47.04 %	47.03 %	0.02 %
Sharan [33]	49.20 %	48.50 %	1.42 %

accurately reproduce key thermodynamic behaviors of the S-CO_2 Brayton cycle.

To further strengthen the validation, detailed comparisons of nodal thermodynamic parameters are performed against Dyreby's benchmark model, as shown in Table 7. The results demonstrate excellent consistency in both pressure and temperature distributions across key system nodes. The maximum deviation is less than 0.002 MPa in pressure and 0.001 °C in temperature.

3.2. Evaluation of the reactor surrogate and mass optimization models

This section evaluates the performance of the developed reactor surrogate model and mass optimization framework. The following subsections analyze the required reactivity margin at BOL, assess the prediction accuracy of the surrogate model, and discuss the results of the reactivity-matching and reactor mass optimization procedures.

3.2.1. Evaluation of reactivity margin at beginning of life

To determine the necessary reactivity margin ensuring long-term reactor criticality, the variation of reactivity decrement (Δk_{eff}) over the entire reactor lifetime is presented in Fig. 8. The dataset comprises a wide range of sample points, each corresponding to a different set of reactor design parameters, but all satisfying the EOL criticality condition ($k_{eff} > 1$).

The results reveal a clear negative correlation between the initial mass of ^{235}U and the observed reactivity loss Δk_{eff}. Specifically, a higher initial fissile fuel inventory significantly reduces the degradation in k_{eff} over the operating lifetime, indicating that increased fuel loading effectively mitigates reactivity degradation caused by fuel depletion.

To guarantee criticality at EOL within the investigated design space—and to incorporate a conservative 5 % safety margin—the required target reactivity at BOL (k_{margin}) is set to 1.26. This benchmark ensures sufficient excess reactivity to support long-term reactor operation and serves as a critical constraint for subsequent reactor design optimization.

3.2.2. Analysis of the prediction results of the surrogate model

To determine the optimal configuration of the reactivity surrogate model, a comprehensive sensitivity analysis is performed on key hyperparameters, including the number of neurons per hidden layer and the choice of activation function.

A two-hidden-layer architecture is employed, with the number of neurons per layer vary from 10 to 50. Each model is trained for 30,000 iterations. The resulting mean squared error (MSE) and coefficient of determination (R^2) are plotted in Fig. 9(a). The analysis indicates that the configuration with 30 neurons per hidden layer yields the best performance, achieving the lowest MSE and an R^2 value approaching 1, thereby confirming strong agreement between model predictions and training data. Consequently, this configuration is selected for the final surrogate model.

Subsequently, the influence of different activation functions is assessed under the optimal neuron configuration. As shown in Fig. 9(b). Although the R^2 values show relatively minor variations across activation types, the Tanh activation function produces the lowest MSE, indicating superior prediction capability compared to other alternatives.

Table 7
Comparison of node parameters from present thermodynamic model and literature.

Node Pressures (MPa)	Literature results	Present models	Node temperature (°C)	Literature results	Present models
p_2	25.7653	25.7653	T_2	64.553	64.552
p_3	25.5076	25.5077	T_3	185.589	185.589
p_4	25.5076	25.5077	T_4	185.295	186.294
p_5	25.2525	25.2526	T_5	358.213	358.213
p_7	8.2449	8.2448	T_7	396.163	396.163
p_8	8.1624	8.1624	T_8	192.070	192.069
$p_{9,10,11}$	8.0808	8.0810	$T_{9,10,11}$	75.834	75.833
p_{12}	25.5076	25.5077	T_{12}	184.822	184.821

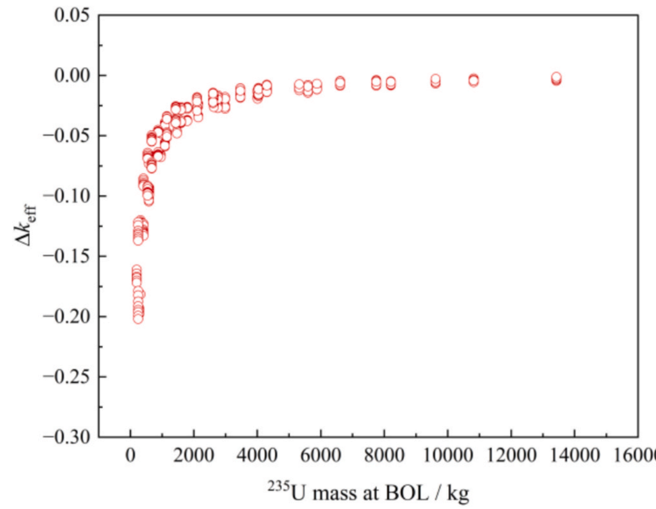


Fig. 8. Distribution of Δk_{eff} over the reactor lifetime for valid design configurations.

To further enhance generalization and prevent overfitting, an early stopping mechanism is integrated into the training process. The maximum number of iterations is set to 50,000, but training is halted earlier when no improvement in validation loss is observed over a predefined number of steps. As shown in Fig. 10, the loss function decreases rapidly during early training and plateaus gradually. Training is automatically terminated at iteration 46,152, ensuring both computational efficiency and model robustness.

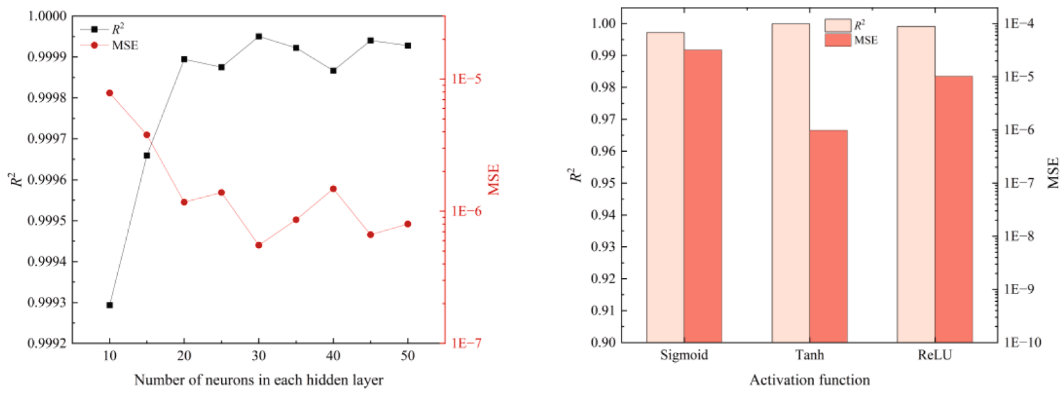
The predictive performance of the trained surrogate model is evaluated using the test dataset. The model achieves an excellent R^2 value of 0.99997 and an MSE of only 3.39×10^{-7}. As depicted in Fig. 11, the predicted k_{eff} values agree well with the reference values, closely following the ideal $y = x$ line. This remarkable agreement demonstrates the surrogate model's high fidelity and accuracy. In addition, the reactivity surrogate model extrapolation risk check is presented in Appendix A.1.2.

In summary, the well-tuned MLP-based surrogate model exhibits excellent fitting performance, generalization capability, and numerical stability, making it a reliable and efficient replacement for computationally expensive Monte Carlo simulations in the subsequent reactivity-matching and mass optimization framework.

3.2.3. Analysis of the matching optimization results of reactivity

Based on the developed neural network surrogate model for reactivity prediction, the model is embedded within the criticality-matching optimization framework to evaluate the performance of different reactor design configurations. A representative case with a fuel volume fraction of 0.24 is first analyzed to examine the influence of reflector thickness δ_{refl} (derived from m_{refl} and R_{core}) on the optimal core radius R_{core} and total reactor mass $M_{Reactor}$.

As shown in Fig. 12, $M_{Reactor}$ exhibits a non-monotonic variation with increasing δ_{refl}— initially decreasing and then rising. When δ_{refl} is relatively small (35.9–43.5 cm), a larger core radius R_{core} is required to meet the reactivity constraint. In this regime, increasing reflector thickness δ_{refl} effectively reduces neutron leakage, thereby lowering the amount of fuel required to achieve k_{margin} and reducing both R_{core} and $M_{Reactor}$. However, as δ_{refl} continues to increase (43.5–51.6 cm), the additional reflector mass outweighs the mass savings from a smaller core, leading to an increase in total mass. Therefore, this opposing trend indicates the existence of an optimal δ_{refl} that minimizes total reactor



(a) (b)

Fig. 9. Sensitivity analysis of surrogate model performance with respect to (a) number of neurons in each hidden layer and (b) activation function.

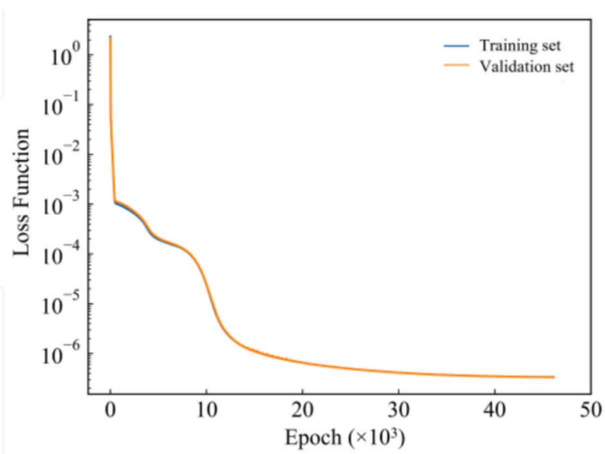


Fig. 10. Training and validation loss evolution.

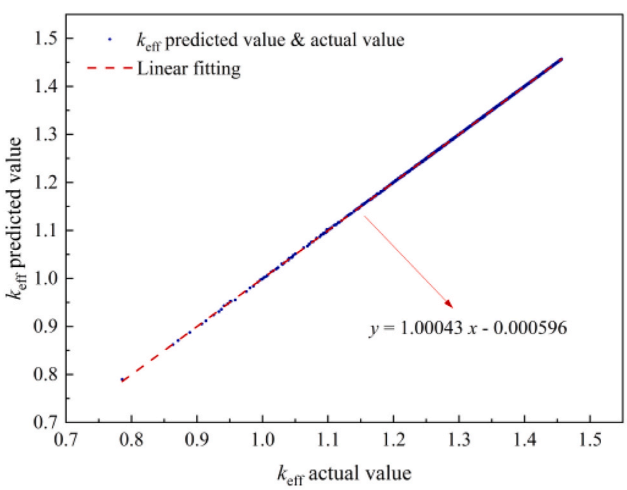


Fig. 11. Comparison of predicted versus true values.

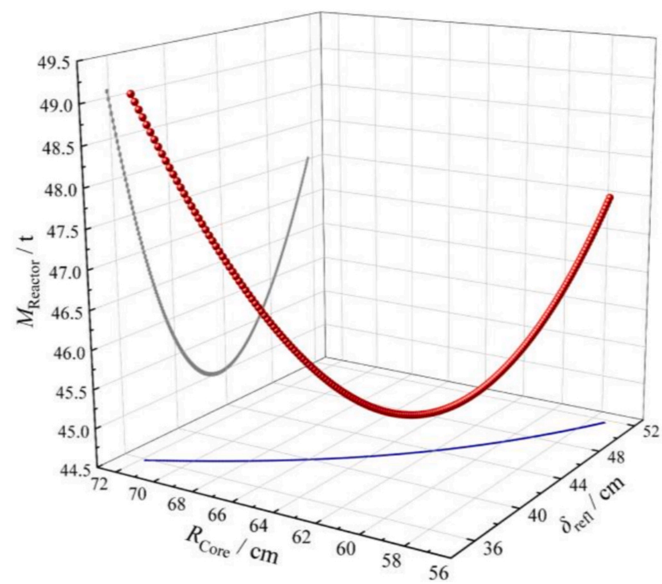


Fig. 12. Reactor mass optimization curve at $f_{\mathrm{fuel}} = 0.24$.

mass while satisfying the critical reactivity margin k_{margin}.

To generalize this observation, the optimization is extended to a design space comprising 36 discrete values of f_{fuel} and 100 values of m_{refl}. For each value of f_{fuel}, the reactor criticality-matching optimization is solved to determine the optimal combination of R_{core}, δ_{refl} and m_{refl} that minimizes the reactor mass M_{Reactor} while satisfying k_{margin}. The resulting relationships are shown in Fig. 13(a) and Fig. 13(b).

The results reveal that with increasing f_{fuel}, R_{core} decreases significantly, while m_{refl} increases overall and δ_{refl} exhibits an overall decreasing trend. In the low-fuel-fraction range ($f_{\mathrm{fuel}} = 0.2-0.4$), higher fuel concentration enables smaller cores but also leads to increased neutron leakage. A moderate increase in m_{refl} is necessary to enhance neutron reflection and maintain reactivity. Nevertheless, due to the more pronounced reduction in R_{core}, δ_{refl} still declines. In the medium range ($f_{\mathrm{fuel}} = 0.4$–0.7), higher values of m_{refl} do not necessarily lead to significant improvement in neutron reflection efficiency. Instead, they may introduce excessive reflector mass, which is unfavorable for overall mass minimization. As a result, the trade-off between reactivity compensation and mass optimization stabilizes in this range, leading to relatively small variations in m_{refl}. In the high range ($f_{\mathrm{fuel}} = 0.7$–0.9), the further increase in fuel concentration leads to further reduction in core radius R_{core}, necessitating larger reflectors to maintain criticality with significant leakage.

The coupling relationships between R_{core}, δ_{refl} and f_{fuel} exhibit distinct nonlinear dependencies. To quantitatively represent these correlations, polynomial regression models of different orders are fitted and compared. Detailed fitting performance evaluations, including verification at non-discrete points (e.g., f_{fuel}=0.21), are provided in Appendix A.1.3. Based on the comparative analysis, a ninth-order polynomial is

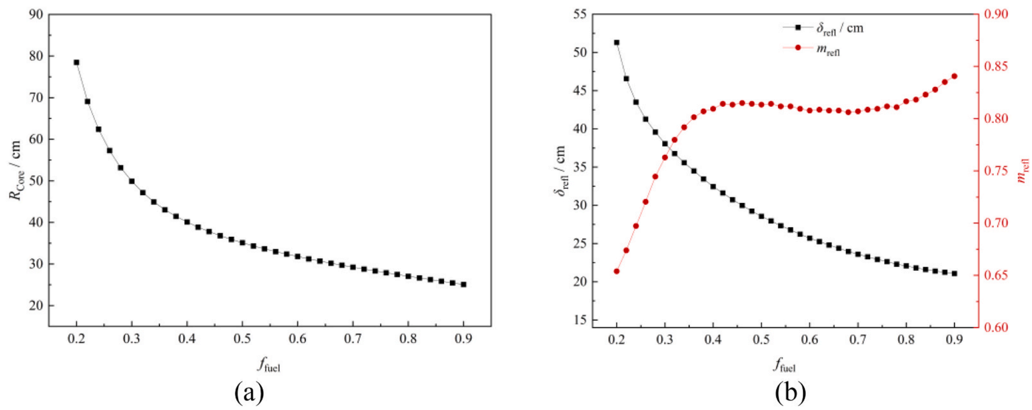


Fig. 13. Relationships between core radius R_{core}, reflector thickness δ_{refl}, reflector thickness multiplier m_{refl} and fuel fraction f_{fuel}: (a) R_{core} vs. f_{fuel} and (b)m_{refl} vs. f_{fuel}.

selected as it most effectively captures the nonlinear interactions among the design parameters, ensuring robustness and numerical stability in the subsequent optimization processes.

3.3. Optimization and decision-making results

This section synthesizes the outcomes of the integrated optimization and decision-making framework. It begins with sensitivity analyses—quantifying the influence of key design parameters —to identify interaction effects. These insights reveal the conflicts among different indicators and establish a rigorous basis for the subsequent multi-objective optimization. Detailed results and comparative plots are provided in Appendix A.2. The following discussion focuses on the convergence characteristics of NSGA-II, the distribution of optimized design variables along the Pareto front, and the final multi-criteria decision-making outcomes obtained via the CRITIC–TOPSIS method, highlighting balanced configurations and their engineering implications.

3.3.1. Optimization results

1) Algorithm parameter performance

To evaluate the robustness and sensitivity of the algorithmic parameters, a systematic parametric study is performed by varying the crossover probability p_c and mutation probability p_m. Specifically, three representative configurations are tested: (p_c, p_m) = (0.95, 0.01), (0.90, 0.10), (0.80, 0.20). A reference point is established based on the worst objective vector in the initial population and further shifts from the objective space to ensure that it is strictly dominated by all subsequent solutions.

Fig. 14 illustrates the evolution of the HV metric for each tested parameter combination. The results reveals that all configurations are capable of recovering high-quality Pareto fronts. However, the (0.95, 0.01) configuration achieves the highest HV values early on, demonstrating the fastest convergence and the most stable trajectory. The (0.90, 0.10) configuration attains a marginally higher final HV but requires more generations and exhibits minor oscillations, while the (0.80, 0.20) configuration shows strong initial exploration but suffers from slower convergence and occasional HV rollback. Overall, the (0.95, 0.01) configuration is recommended for optimal performance, providing a robust trade-off between speed, stability, and computational cost.

2) Distribution of optimized variables

To gain deeper insights into the interdependence between conflicting objectives and their corresponding design parameters after optimization, the distribution characteristics of decision variables along the Pareto front are analyzed. Fig. 15 presents the scatter distributions of key variables within the Pareto-optimal set.

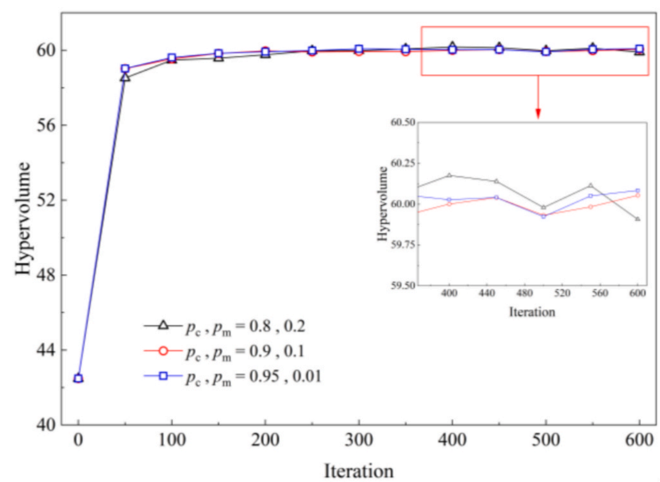


Fig. 14. Evolution of the HV metric for each tested parameter configuration.

As shown in Fig. 15(a), the optimized main-compressor inlet pressure p_1 clusters tightly within 7.64–7.68 MPa, while the turbine inlet pressure p_6 tends toward the upper bound. This indicates a clear preference for higher p_6, which enhances turbine work and thereby improves both thermal efficiency and specific work, with only a marginal penalty in system mass. In contrast, p_1 remains constrained within a narrow range to avoid penalizing any objective.

In Fig. 15(b), the total heat conductance UA_{Total} remains close to its lower bound, revealing that the optimizer deliberately relaxes heat-transfer performance to reduce the heat exchanger area and system mass. The split ratio x concentrates between 0.10 and 0.35, avoiding extreme values and maintaining balanced mass-flow allocation that preserves recuperation effectiveness.

Fig. 15(c) shows that the conductance allocation ratios (φ_{LTR} and φ_{HTR}) predominantly fall between 0.30 and 0.35, indicating a balanced distribution among recuperators and the cooler rather than over-reinforcing a single unit.

Finally, as depicted in Fig. 15(d), the mass fluxes G_i of the heat exchanger cluster around 900 kg/(m^2·s). While further increasing G_i has limited effect on efficiency or specific work, it effectively reduces mass. Hence, the optimization process naturally drives the G toward the upper bound to minimize total system mass.

Overall, these distributions demonstrate that the multi-objective optimization framework effectively balances the trade-offs among thermal efficiency, specific work, and total mass, revealing coherent physical trends consistent with engineering intuition.

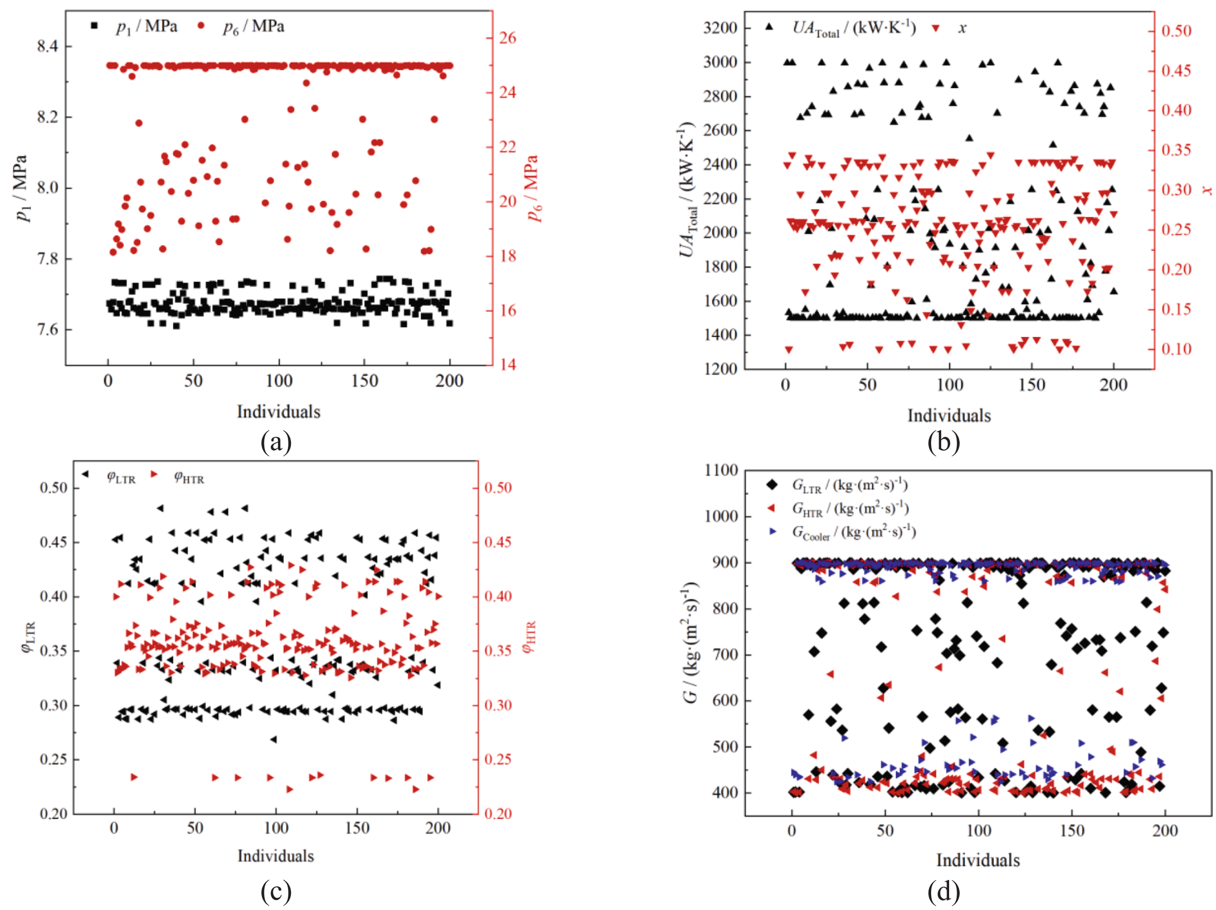


Fig. 15. Distribution of decision variables along the Pareto front: (a) p_1 & p_6; (b) UA_{Total} & x; (c) φ_{LTR} & φ_{HTR} and (d).G_{LTR} & G_{HTR} & G_{Cooler}

3.3.2. *Decision making results*

To identify the optimal reactor–power cycle configuration, a multi-criteria decision analysis is conducted by integrating the CRITIC weighting method and the TOPSIS decision-making framework. Using the CRITIC method, the objective weights are derived based on the variability and conflict among three performance metrics (thermal efficiency, specific work, and system mass). As shown in Fig. 16, the calculated weights gradually converge and stabilize as the sample size increases, indicating consistent objective sensitivity. The final weight distribution reveals that system mass (42.67 %) holds the highest importance, followed by thermal efficiency (30.33 %) and specific work (27.00 %), reflecting the engineering priority toward lightweight system design.

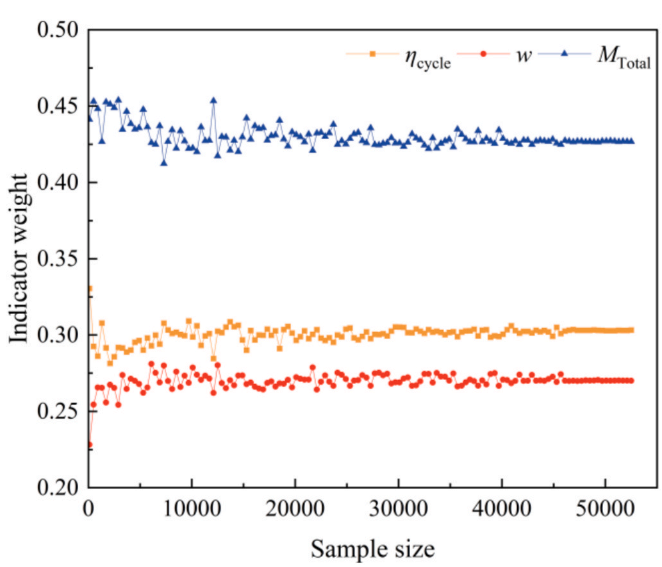


Fig. 16. Calculation results of the objective weights by the CRITIC method.

Based on the obtained weights, the TOPSIS method is applied to the Pareto-optimal solutions to determine the most balanced configuration. The decision outcomes are presented in Fig. 17, while the detailed ranking result is summarized in Appendix A.3. It can be observed that the performance variation among the top-ranked solutions is minimal—thermal efficiency varies by less than 1.5 percentage points, specific work by approximately 16 kJ/kg, and total mass by less than 1.2 tons. These small variations indicate that the top-ranked configurations exhibit comparable overall performance and demonstrate strong convergence toward balanced multi-objective designs, rather than being dominated by any single performance criterion.

A comparison between the TOPSIS-optimal configuration and the individual objective-optimal solutions is presented in Table 8. Due to the dominant weight of system mass M_{Total} (accounting for the largest proportion among the three objectives), the parameter values of the TOPSIS-optimal design tend to align more closely with those of the minimum-mass configuration. This is particularly evident in the values of heat conductance UA_{Total} and heat exchanger mass fluxes G_{LTR}、G_{HTR} and G_{Cooler}.

Compared with the configurations that maximize either thermal efficiency or specific work, the TOPSIS-optimal design sacrifices 9.53 % in efficiency and 7.75 % in specific work, respectively, but achieves substantial reductions of 48.62 % and 21.03 % in system mass, respectively,

Such a trade-off demonstrates that modest sacrifices in thermodynamic performance can yield significant benefits in compactness. The

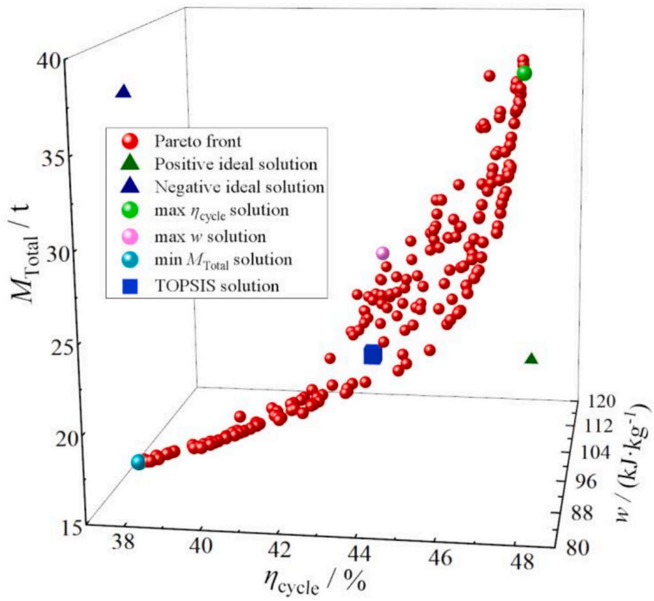


Fig. 17. Optimization and decision-making results using the CRITIC–TOPSIS method.

resulting mass reduction mitigates challenges related to transportation, structural loading, and installation, offering particular advantages in marine, mobile, and modular nuclear applications. By strategically balancing thermodynamic performance and mass, the proposed approach provides a practical decision-making pathway for next-generation S-CO_2 nuclear systems, where lightweight, high-efficiency, and robust coupling are essential.

To further demonstrate and compare the outcomes of optimization and decision-making, Fig. 18 illustrates the component-wise system mass distribution under different decision strategies. The TOPSIS-optimal configuration achieves a substantial reduction in system mass relative to the configurations optimized solely for maximum cycle efficiency η_{cycle} or maximum specific work w. This improvement is primarily driven by notable reductions in the masses of the LTR and HTR. In contrast, the masses of the reactor and the cooler exhibit relatively minor variations among different decision strategies.

Following the selection of the TOPSIS-optimal configuration, it is reintegrated into both the S-CO_2 thermodynamic model and the reactor mass evaluation model to perform comprehensive system node and reactor parameter analyses. The corresponding thermodynamic states of the working fluid at key system nodes and the detailed reactor structural parameters are summarized in Table 9 and Table 10, respectively.

4. Conclusion

This study develops a comprehensive integrated framework to modeling, performance analysis, multi-objective optimization, and decision-making of S-CO_2 nuclear energy systems, with emphasis on the integrated coupling between thermodynamic cycle performance and reactor design characteristics. A unified and detailed system-level methodology is established to evaluate critical trade-offs among thermal efficiency, specific power, and system mass, and to guide optimal design decisions. The key findings and contributions are summarized as follows:

(1) A thermodynamic model of the S-CO_2 recompression Brayton cycle is developed and validated against benchmark literature data, confirming its accuracy and reliability for predicting key performance indicators such as efficiency and node thermodynamic states.
(2) A reactor mass model for the S-CO_2 nuclear reactor is constructed by considering fuel inventory, core geometry, reflector thickness, and pressure vessel design. Through burnup and reactivity margin analyses, by incorporating burnup calculations and establishing the required initial excess reactivity margin, a physically grounded reactivity constraint is imposed to ensure sustained criticality across the entire core lifetime. To accelerate the core design evaluation process, a neural network surrogate model is constructed to capture the nonlinear relationships between design variables and reactivity response, achieving a

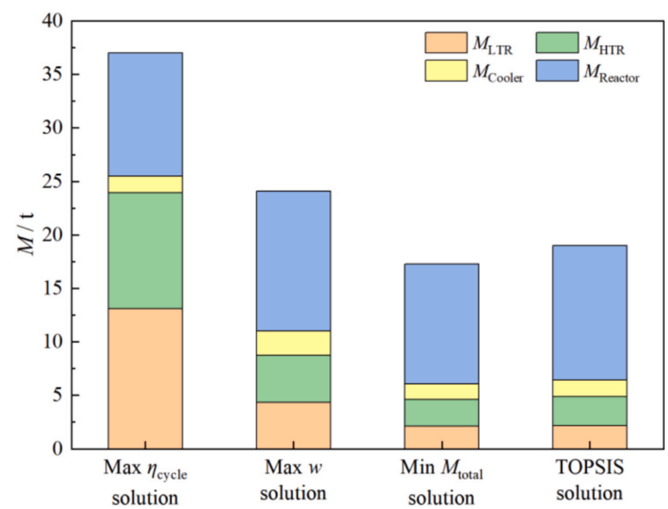


Fig. 18. Component-wise mass distribution of different decision-making solutions.

Table 8
Comparison of decision-making results.

Parameters		Max η_{cycle} solution	Max w solution	Min M_{Total} solution	TOPSIS solution
Optimized variables	p_1/MPa	7.734	7.658	7.678	7.659
	p_6/MPa	24.999	24.999	18.146	24.854
	UA_{Total}/(kW/K)	2996.768	1533.364	1504.034	1503.237
	x	0.345	0.100	0.261	0.219
	φ_{LTR}	0.454	0.339	0.289	0.297
	φ_{HTR}	0.412	0.330	0.334	0.362
	G_{LTR}/[kg/(m^2·s)]	402.397	402.639	899.224	890.809
	G_{HTR}/[kg/(m^2·s)]	401.763	400.580	897.682	862.510
	G_{Cooler}/[kg/(m^2·s)]	434.938	439.884	899.313	891.513
Optimized objectives	η_{cycle}/%	47.524	42.892	37.940	42.995
	w/(kJ/g)	108.787	122.029	86.158	112.572
	M_{Total}/t	37.029	24.091	17.270	19.024
Decision making score	C_i/10^{-2}	20.305	67.558	76.131	87.039

Table 9
System node thermodynamic parameters under the TOPSIS solution.

Working medium	Node	p/MPa	T/°C	Working medium	Node	p/MPa	T/°C
CO_2	1	7.659	32.000	CO_2	7	7.846	409.313
	2	25.398	70.737		8	7.744	177.537
	3	25.389	140.714		9	7.690	83.708
	4	25.389	152.711		10	7.690	83.708
	5	25.361	341.050		11	7.690	83.708
	6	24.854	550.000		12	25.389	200.666
Cooling water	13	0.500	20.000	Cooling water	14	0.404	35.865

Table 10
Reactor design parameters under the TOPSIS solution.

Reactor parameters	Value	Reactor parameters	Value
f_{fuel}	0.479	N_{cool}	2381
R_{core}/cm	35.922	HD_{core}	1.5
r_{cool}/cm	0.5	M_{core}/t	2.547
δ_{clad}/cm	0.031	M_{refl}/t	5.365
δ_{refl}/cm	29.256	M_{pv}/t	4.644
δ_{pv}/cm	5.626	$M_{Reactor}$/t	12.556

prediction accuracy of $R^2 = 0.99997$. This surrogate enables rapid reactor mass optimization and reveals nonlinear coupling behaviors among fuel fraction, core radius, and reflector thickness.

(3) A systematic sensitivity analysis is performed to evaluate the influence of key design variables—including main compressor inlet pressure, turbine inlet pressure, total heat conductance, mass flow split ratio, conductance distribution ratio, and heat exchanger mass flux—on system performance as presented in Appendix A.2. The analysis reveals non-monotonic behaviors and interdependent trends, indicating the existence of optimal parameter regions. Moreover, distinct trade-offs are observed among thermal efficiency, specific work, and total system mass, offering valuable guidance for system-level optimization.

(4) A multi-objective optimization of the integrated $S\text{-}CO_2$ nuclear energy system is conducted using the NSGA-II genetic algorithm, targeting the simultaneous improvement of thermal efficiency, specific power, and total system mass. To systematically extract the balanced configuration from the Pareto front, a CRITIC–TOPSIS decision-making method is employed. The CRITIC method assigns objective weights based on statistical variability and correlation, identifying system mass as the most influential objective (42.67 %), followed by thermal efficiency (30.33 %) and specific work (27.00 %). Compared with the configurations that maximize either thermal efficiency or specific work, the TOPSIS method identifies a balanced solution that achieves over 48.62 % and 21.03 % mass reduction while only sacrifices 9.53 % in efficiency and 7.75 % in specific work, respectively, demonstrating effective compromise among conflicting objectives.

(5) Detailed analysis of the TOPSIS-optimal solution provides a complete system configuration, including pressure–temperature states at all thermodynamic nodes and core structural parameters. The final optimized design features a compact reactor with a core radius of 35.922 cm, a reflector thickness of 29.256 cm, and a total reactor mass of 12.556 t—successfully balancing thermal performance, mass, and reactivity.

In summary, this work establishes a unified methodology that integrates physics-based modeling, surrogate-assisted optimization, and objective decision analysis for the design of $S\text{-}CO_2$ nuclear energy systems. The proposed approach provides valuable theoretical guidance and practical references for advanced $S\text{-}CO_2$ nuclear energy systems, supporting future engineering applications and deployment. In addition, the proposed methodology is generalizable and can also be extendable to non-nuclear compact $S\text{-}CO_2$ systems, such as solar-thermal or waste-heat recovery applications. Future development will focus on:

(1) Extending the current steady-state framework toward transient-aware surrogate modeling to capture load-following and dynamic responses. This enhancement will enable the framework to handle non-steady conditions and improve its applicability to realistic reactor operating scenarios.
(2) Incorporating experimental or industrial validation data for $S\text{-}CO_2$ nuclear systems to enhance the model's predictive robustness, reliability and real-world applicability.

CRediT authorship contribution statement

Yu Shen: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Xiafeng Zhou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.enconman.2025.120852.

Data availability

Data will be made available on request.

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